

Import Packages

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
In [ ]: # Access to google drive
from google.colab import drive
drive.mount('/content/drive')
```

Load Data

Load data from the Github repository. Examples are below.

```
In [2]: # Load data without column names
ExampleData = pd.read_csv('https://github.com/purdueIamm/purdue_me597_iiot/blob/main/ml_tutorial/ML1/ExampleData')
ExampleData
```

```
Out[2]:
```

	0	1	2	3
0	0.000000	0.016440	-0.014200	0.001433
1	0.000078	0.013822	-0.015707	0.001504
2	0.000156	0.019431	-0.016963	0.001433
3	0.000234	0.003726	-0.017248	0.001549
4	0.000313	0.022049	-0.009840	0.001864
...	...	...	...	...
2769	0.216330	0.077392	0.002404	0.003453
2770	0.216410	0.030276	0.003952	0.003498
2771	0.216480	0.017188	0.005504	0.003397
2772	0.216560	0.037754	-0.000108	0.003506
2773	0.216640	-0.016093	-0.001652	0.003600

2774 rows × 4 columns

```
In [3]: # Load data with column names
path = 'https://github.com/purdueIamm/purdue_me597_iiot/blob/main/ml_tutorial/ML1/ExampleData?raw=true' # yo
ExampleData_withNames = pd.read_csv(path, sep=',', names=['time(s)', 'Acceleration(g)', 'Voltage(V)', 'Curren
ExampleData_withNames
```

```
Out[3]:
```

	time(s)	Acceleration(g)	Voltage(V)	Current(kA)
0	0.000000	0.016440	-0.014200	0.001433
1	0.000078	0.013822	-0.015707	0.001504
2	0.000156	0.019431	-0.016963	0.001433
3	0.000234	0.003726	-0.017248	0.001549
4	0.000313	0.022049	-0.009840	0.001864
...	...	...	...	...
2769	0.216330	0.077392	0.002404	0.003453
2770	0.216410	0.030276	0.003952	0.003498
2771	0.216480	0.017188	0.005504	0.003397
2772	0.216560	0.037754	-0.000108	0.003506
2773	0.216640	-0.016093	-0.001652	0.003600

2774 rows × 4 columns

Tip: You can try other ways to load data in Colab by referring to:

<https://towardsdatascience.com/4-awesome-ways-of-loading-ml-data-in-google-colab-9a5264c61966>

(4 Awesome Ways Of Loading ML Data In Google Colab)

Data Handling

Define only sensor signals without time information

= Extract a part of ExampleData from second(1) column to last(3) column

```
In [4]: SensorDataOnly = ExampleData.iloc[:,1:]
SensorDataOnly
```

```
Out[4]:
```

	1	2	3
0	0.016440	-0.014200	0.001433
1	0.013822	-0.015707	0.001504
2	0.019431	-0.016963	0.001433
3	0.003726	-0.017248	0.001549
4	0.022049	-0.009840	0.001864
...	...	...	...
2769	0.077392	0.002404	0.003453
2770	0.030276	0.003952	0.003498
2771	0.017188	0.005504	0.003397
2772	0.037754	-0.000108	0.003506
2773	-0.016093	-0.001652	0.003600

2774 rows × 3 columns

Extract data from 0.01 seconds to 0.02 seconds

```
In [5]: StartPoint = np.where(ExampleData.iloc[:,0].values == 0.01)[0][0]
        EndPoint   = np.where(ExampleData.iloc[:,0].values == 0.02)[0][0]

        StartPoint, EndPoint
```

```
Out[5]: (np.int64(128), np.int64(256))
```

```
In [6]: ExampleData_withNames.iloc[128:256+1,:]
```

```
Out[6]:
```

	time(s)	Acceleration(g)	Voltage(V)	Current(kA)
128	0.010000	-0.017962	-0.010766	-0.005784
129	0.010078	-0.094246	0.022769	0.001770
130	0.010156	-0.103970	-0.086215	-0.008154
131	0.010234	-0.103590	0.165990	0.005673
132	0.010313	-0.103220	-0.925800	-0.033911
...	...	...	...	...
252	0.019688	-0.213910	1.954200	3.787500
253	0.019766	-0.256910	1.879100	3.767100
254	0.019844	-0.278220	1.790100	3.738200
255	0.019922	-0.379560	1.695900	3.700200
256	0.020000	-0.400500	1.602600	3.650400

129 rows × 4 columns

```
In [7]: SensorSignalOnly = ExampleData_withNames.iloc[StartPoint:EndPoint, :]
        SensorSignalOnly
```

```
Out[7]:
```

	time(s)	Acceleration(g)	Voltage(V)	Current(kA)
128	0.010000	-0.017962	-0.010766	-0.005784
129	0.010078	-0.094246	0.022769	0.001770
130	0.010156	-0.103970	-0.086215	-0.008154
131	0.010234	-0.103590	0.165990	0.005673
132	0.010313	-0.103220	-0.925800	-0.033911
...	...	...	...	...
251	0.019609	-0.222880	2.025500	3.797800
252	0.019688	-0.213910	1.954200	3.787500
253	0.019766	-0.256910	1.879100	3.767100
254	0.019844	-0.278220	1.790100	3.738200
255	0.019922	-0.379560	1.695900	3.700200

128 rows × 4 columns

Transpose the matrix of data

```
In [8]: TransposedData = SensorSignalOnly.T
        TransposedData
```

Out[8]:

	128	129	130	131	132	133	134	135	136	
time(s)	0.010000	0.010078	0.010156	0.010234	0.010313	0.010391	0.010469	0.010547	0.010625	0.010
Acceleration(g)	-0.017962	-0.094246	-0.103970	-0.103590	-0.103220	-0.104720	0.010083	-0.015719	-0.043390	0.172
Voltage(V)	-0.010766	0.022769	-0.086215	0.165990	-0.925800	-2.144800	-1.772900	-1.719100	-1.729600	-1.665
Current(kA)	-0.005784	0.001770	-0.008154	0.005673	-0.033911	-0.274470	-0.533520	-0.746910	-0.964190	-1.147

4 rows × 128 columns



Save data

Save data as a file in your GoogleDrive

```
In [10]: path = '../data/Data_ML1_1.csv'
SensorSignalOnly.to_csv(path, sep=',', index=None)
```

Load the data you just saved in your GoogleDrive

```
In [11]: SensorSignalOnly_load = pd.read_csv(path)
SensorSignalOnly_load
```

Out[11]:

	time(s)	Acceleration(g)	Voltage(V)	Current(kA)
0	0.010000	-0.017962	-0.010766	-0.005784
1	0.010078	-0.094246	0.022769	0.001770
2	0.010156	-0.103970	-0.086215	-0.008154
3	0.010234	-0.103590	0.165990	0.005673
4	0.010313	-0.103220	-0.925800	-0.033911
...	...	...	...	...
123	0.019609	-0.222880	2.025500	3.797800
124	0.019688	-0.213910	1.954200	3.787500
125	0.019766	-0.256910	1.879100	3.767100
126	0.019844	-0.278220	1.790100	3.738200
127	0.019922	-0.379560	1.695900	3.700200

128 rows × 4 columns

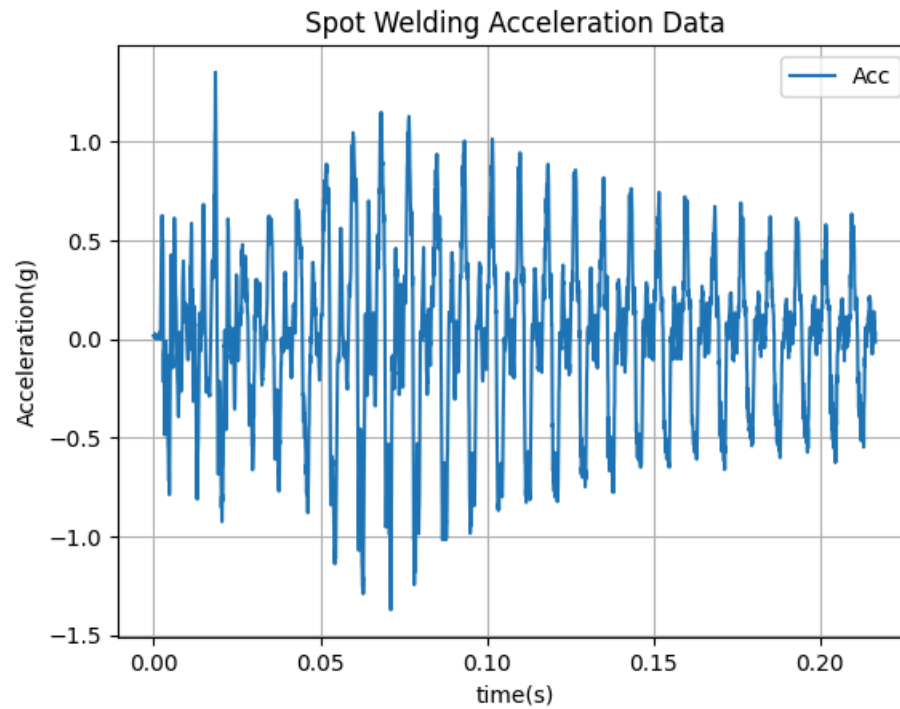
Plot Graphs

Graph options

```
In [12]: # ExampleData Column 0: time
# ExampleData Column 1: Acceleration signal

plt.plot(ExampleData.iloc[:,0], ExampleData.iloc[:,1])
plt.grid()
plt.xlabel('time(s)')
plt.ylabel('Acceleration(g)')
plt.title('Spot Welding Acceleration Data')
plt.legend(['Acc'], loc = 'upper right', fontsize=10)
# plt.xlim(0,0.02)
```

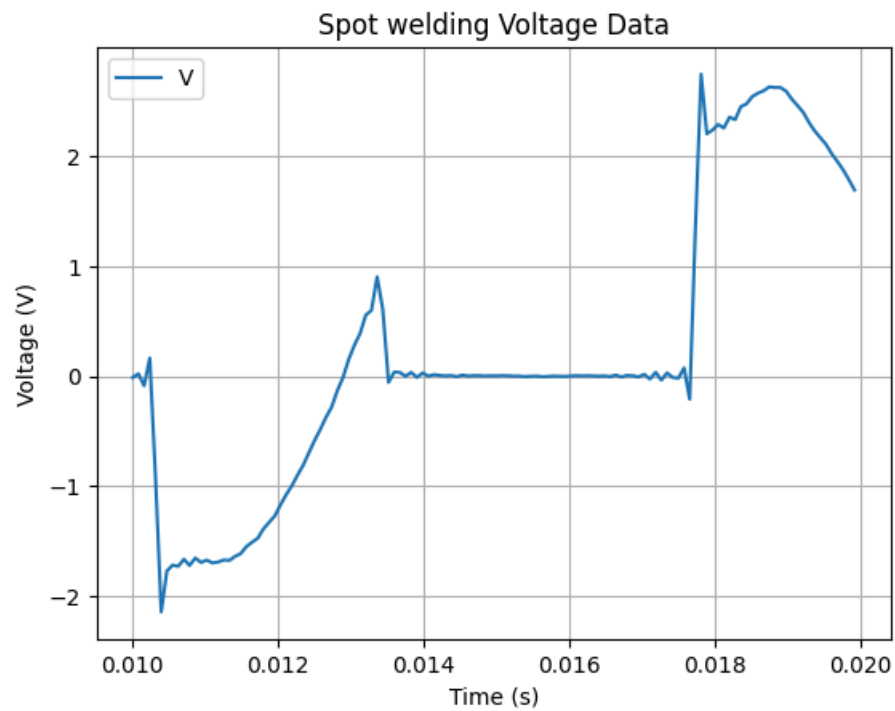
```
# plt.ylim(-1.5,1.5)
plt.show()
```



```
In [16]: # ExampleData Column 0: time
# ExampleData Column 2: Voltage signal

# ylabel: Voltage(V)
# title: Spot welding Voltage Data
# Legend: Voltage (located at 'upper center')

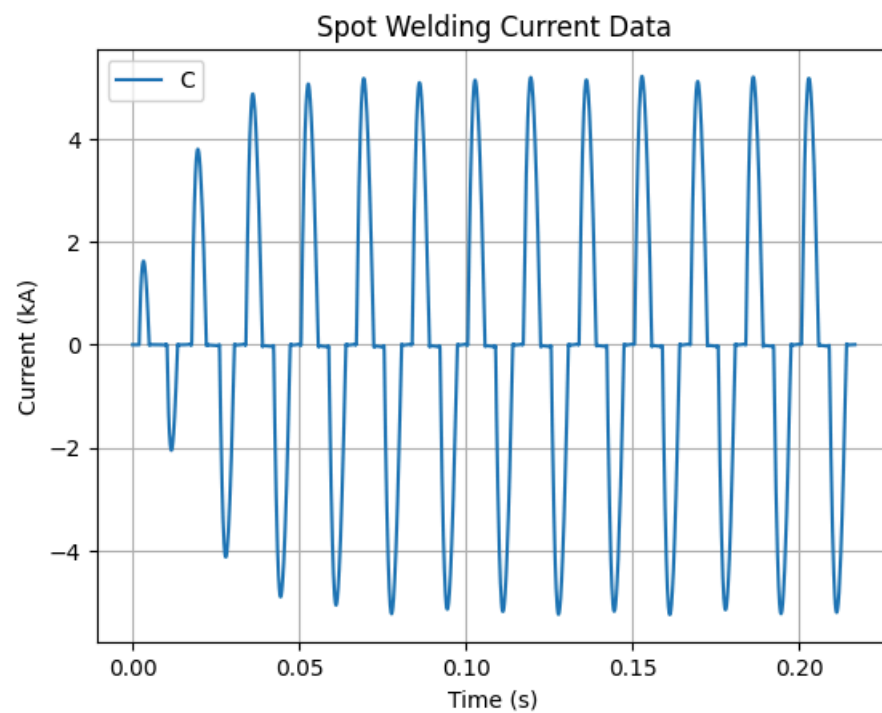
plt.plot(SensorSignalOnly['time(s)'], SensorSignalOnly['Voltage(V)'])
plt.grid()
plt.xlabel("Time (s)")
plt.ylabel("Voltage (V)")
plt.title("Spot welding Voltage Data")
plt.legend("Voltage (located at 'upper center')")
plt.show()
```



```
In [17]: # ExampleData Column 0: time
# ExampleData Column 3: Current signal

# ylabel: Current(kA)
# title: Spot Welding Current Data
# Legend: Current (located at 'best')

plt.plot(ExampleData.iloc[:,0], ExampleData.iloc[:,3])
plt.grid()
plt.xlabel("Time (s)")
plt.ylabel("Current (kA)")
plt.title("Spot Welding Current Data")
plt.legend("Current (located at 'best')")
plt.show()
```

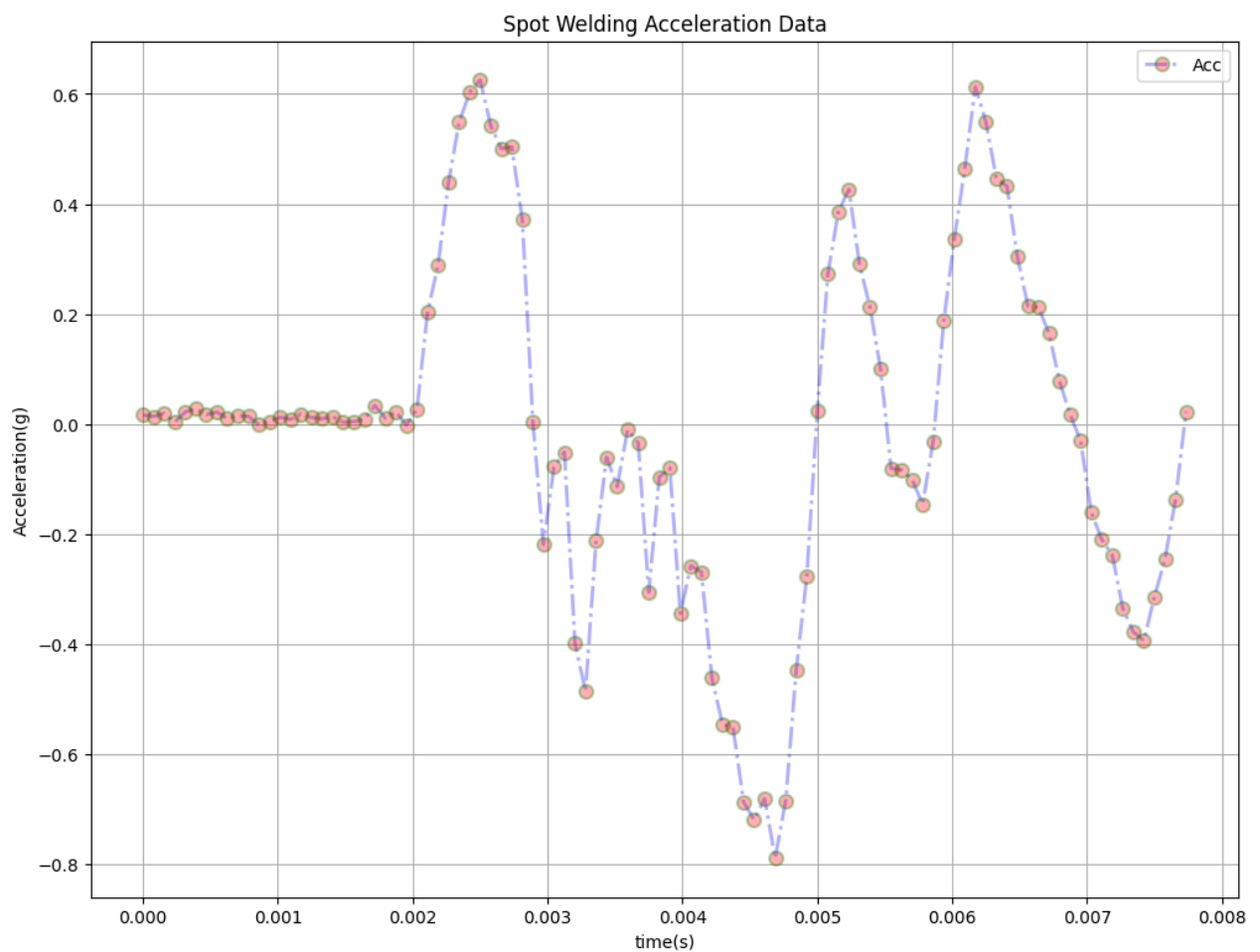


Graph shapes

```
In [18]: plt.figure(figsize = (12,9))                                # figure size
plt.plot(ExampleData.iloc[:100,0],ExampleData.iloc[:100,1],        # data (x, y)
         linestyle = '-.',                                         # line style
         linewidth  = 2.0,                                         # line width
         color      = 'b',                                         # line color
         marker     = 'o',                                         # marker shape
         markersize  = 8,                                         # marker size
         markeredgecolor = 'g',                                    # m.edge color
         markeredgewidth = 1.5,                                   # m.edge width
         markerfacecolor = 'r',                                   # m.face color
         alpha      = 0.3)                                         # opacity

plt.grid()
plt.xlabel('time(s)')
plt.ylabel('Acceleration(g)')
plt.title('Spot Welding Acceleration Data')
plt.legend(['Acc'], loc = 'best', fontsize=10)
```

Out[18]: <matplotlib.legend.Legend at 0x215804ba710>



Combine multiple plots

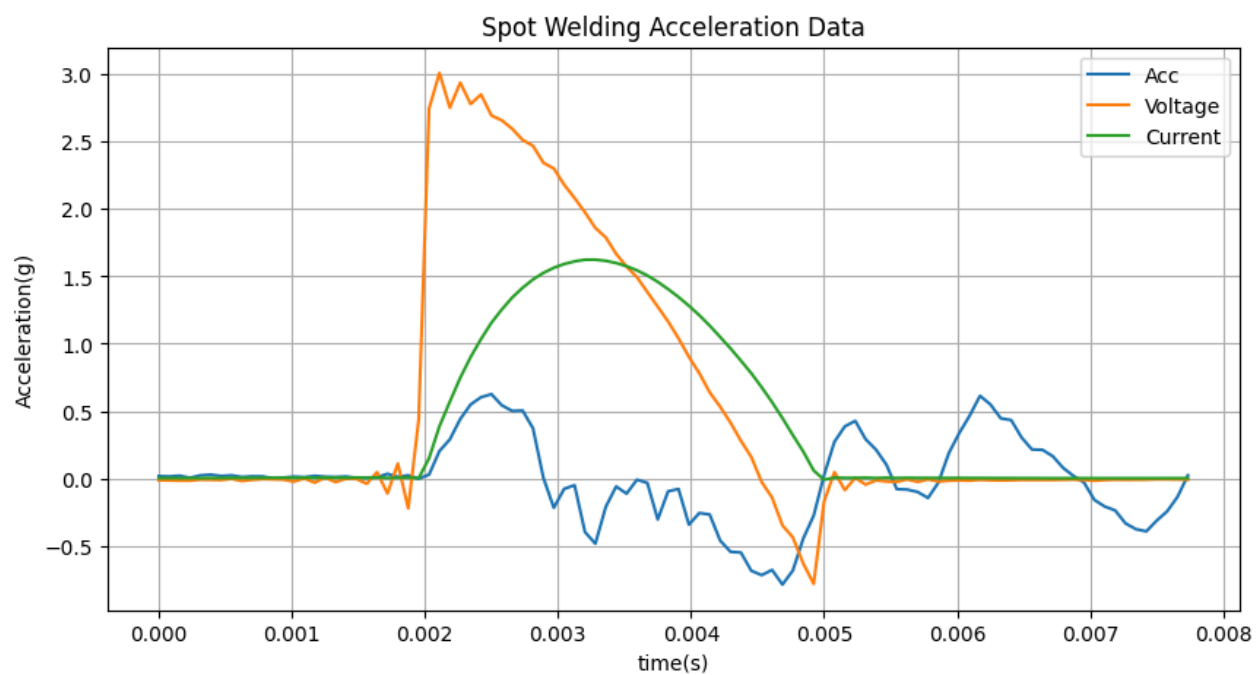
```
In [19]: DataLength = 100 # Define plot Length (~ DataLength/12800 seconds)

plt.figure(figsize = (10,5))
plt.plot(ExampleData.iloc[:DataLength,0],ExampleData.iloc[:DataLength,1])
plt.plot(ExampleData.iloc[:DataLength,0],ExampleData.iloc[:DataLength,2])
plt.plot(ExampleData.iloc[:DataLength,0],ExampleData.iloc[:DataLength,3])

plt.xlabel('time(s)')
plt.ylabel('Acceleration(g)')
```

```
plt.title('Spot Welding Acceleration Data')
plt.legend(['Acc', 'Voltage', 'Current'], loc = 'upper right', fontsize=10)

plt.grid()
plt.show()
```



In []: